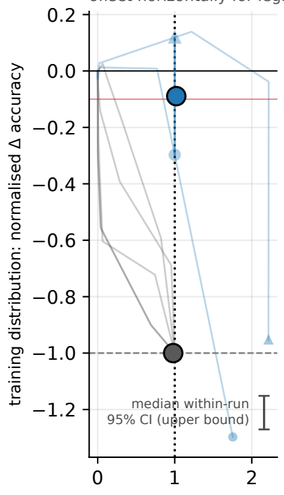
Repair methods at matched hack removal (R = 1 on the training distribution); 5 runs: Gemma s2, Gemma s3, Qwen s1, Qwen s2, Qwen s3
Capability change is normalised by each run's RL gain on the training distribution (0: hacked anchor; −1: untrained)

A. Capability cost on the training distribution

Optimal: maximal y at R = 1 (hack removed, capability retained)

Faint lines: per-run dose curves from the hacked anchor (0, 0). Means at R = 1 are offset horizontally for legibility.

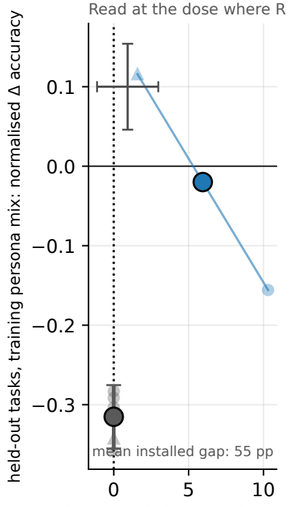


R (fraction of the installed hack removed on the training distribution; R = 1: untrained rate)

B. Removal on held-out tasks

Optimal: x = 0 with maximal y

Read at the dose where R = 1 on the training distribution.

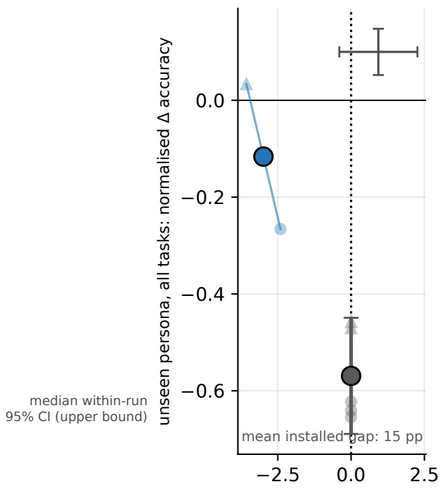


creature rate − untrained rate, held-out tasks, training persona mix (pp)
0: untrained rate; > 0: residual hack; < 0: below untrained

C. Removal under an unseen pers

Optimal: x = 0 with maximal y

Read at the dose where R = 1 on the training

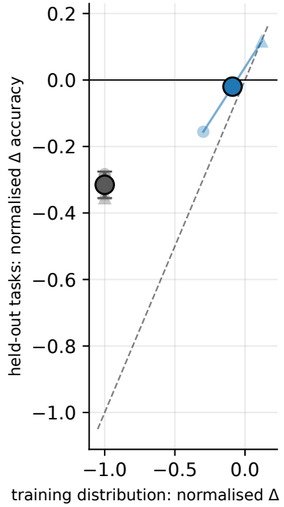


creature rate − untrained rate, unseen persona, all tasks (pp)
0: untrained rate; > 0: residual hack; < 0: below untrained

D. Where the capability cost falls

Optimal: upper right (no cost on either)

Read as in B. Above the diagonal: less cost on held-out tasks.

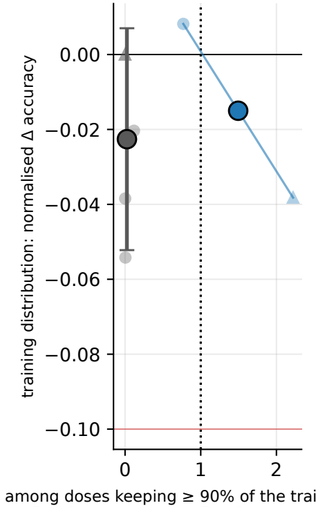


training distribution: normalised Δ accuracy

E. Largest removal keeping 90% of the RL gain

x ≥ 1: full removal within the capability budget

Sampled doses only, no interpolation; means over all runs. No feasible dose, placed at the anchor: rewind to a checkpoint in 2 runs.



largest R among doses keeping ≥ 90% of the training-distribution gain
R < 1: residual hack; R > 1: below untrained

Filled markers: mean over runs reaching R = 1;
error bars: 95% t-interval across runs (panel E: all runs).

- rewind to a checkpoint (R = 1 in 5/5 runs)
- reverse (R = 1 in 2/2 runs)
- individual run, Qwen
- individual run, Gemma
- individual run not reaching R = 1
- mean over fewer than half of runs